Deden Hidayat

Software Engineer · Full-Stack & Industrial IoT

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SUMMARY

Software engineer who builds systems from the factory floor to the cloud — custom hardware and ESP32 firmware, industrial protocol middleware (10+ protocols), real-time MQTT pipelines, and Next.js control centers. I've wired the panels, written the firmware, and shipped the dashboards, so I design for the whole chain. I care about systems that hold up in production, not only in a demo. Open to remote engineering roles and freelance.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C / C++, Dart
Frontend: Next.js, React, Vue 3, Tailwind CSS
Backend: Node.js, Flask, FastAPI, Prisma, PostgreSQL, Supabase, REST & WebSocket APIs
IoT / Protocols: MQTT, Modbus RTU/TCP, BACnet/IP, SNMP, OPC-UA, EtherNet/IP, PROFINET, GOOSE (IEC 61850), IEC 104, Sparkplug B
Embedded: ESP32, Raspberry Pi, NanoPi, ESP-IDF, Arduino, RS485
ML / Data: scikit-learn, LightGBM, predictive maintenance (RUL, fault detection)
Tools / Infra: Docker, systemd, Git, Linux, GitHub Actions, ROS2, Vercel

PROFESSIONAL EXPERIENCE

IoT Developer — PT Graha Sumber Prima Elektronik (GSPE) Jan 2024 – Present
West Jakarta, Indonesia

- Built industrial protocol middleware that exposes one shared data source across 11 protocols (Modbus, BACnet, SNMP, PROFINET, GOOSE, OPC-UA, IEC 104, Sparkplug) to SCADA / BMS / NMS systems, deployed via systemd.
- Built firmware and software for an ESP32 access controller — one controller driving up to 10 locks over RS485/Modbus, reachable over embedded web UI, WebSocket, and MQTT at once.
- Shipped a fleet OTA control plane (CI → GHCR releases, staged rollout, health-checked auto-rollback) so remote devices update without site visits.
- Built real-time MQTT/WebSocket telemetry pipelines feeding SMARTRACK, a full-stack DCIM control center for power, cooling, and multi-protocol IoT.
- Built AI/ML systems for production monitoring — a Vision-LLM machine-activity monitor and a predictive-maintenance engine (bearing-fault, remaining-useful-life, battery SoH) served as a Docker sidecar.

SELECTED PROJECTS

Protocol OUT — Industrial Protocol Gateway · *Python, C, systemd*
One gateway serving 11 industrial protocols in parallel from a single source. PROFINET on a native C core; GOOSE over raw L2 multicast. Deployed as a production systemd service.

LockAccessController — Firmware · Full-Stack · *ESP32, C++, ESP-IDF, Modbus/RS485, MQTT*
ESP32 access-control system (firmware + software). One controller acts as a Modbus master to up to 10 locks over RS485, reachable over embedded web UI, WebSocket, and MQTT at once, with offline-safe logging (RTC + EEPROM).

OTA Central — Fleet Update Control Plane · *Next.js, Prisma, GitHub Actions, GHCR, Docker*
Pull-based OTA for remote devices — staged rollout (10→50→100%), health-check, auto-rollback. Updates remote devices with zero site visits.

ML Predictive-Maintenance Engine · *Python, FastAPI, scikit-learn, LightGBM*
Stateless ML sidecar for an industrial IoT platform. Trained models for bearing-fault (CWRU), remaining-useful-life (FEMTO), and battery SoH; degraded rule-based fallback so the app never hard-fails.

SMARTRACK — DCIM Platform · *Next.js, TypeScript, Prisma, MQTT, Docker* · **Live demo**
Full-stack data-center management platform with real-time MQTT/WebSocket monitoring, multi-protocol IoT ingest, RBAC + AES-256, and PUE/carbon tracking. → smartrack-software.vercel.app

EDUCATION

SMK Negeri Jawa Tengah — Teknik Elektronika Industri 2020 – 2023

- Industrial electronics: panel schematics, PCB design, microcontrollers, and hardware-software integration — the foundation for how I approach hardware-software systems today.